

Detailed Data Mining Project Plan

Categorization of Residential Electricity Consumption Patterns Using Clustering and Classification

Recommended dataset: Electricity Smart Meter Data from Indian Two Cities (Mathura and Bareilly, Uttar Pradesh)

Objective of this document: Explain the dataset, the attributes, the data transformation, the Data Mining methods, evaluation plan, expected outputs, and the direct connection to 23CS3551.

1. Problem Statement

Problem: Residential consumers have different electricity-usage behaviors. The raw smart-meter readings contain these patterns, but the consumer types are not explicitly labelled.

Proposed solution: Discover groups of meters with similar electricity-consumption behavior using clustering, interpret those groups, and optionally train supervised models to classify an unseen consumption profile into one of the discovered groups.

Research question: Can historical electricity-consumption data be mined to discover meaningful residential consumption profiles and reliably recognize those profiles for unseen data?

2. Why We Chose This Dataset

Reason	Why it matters
Indian data	Smart meters are from Mathura and Bareilly, Uttar Pradesh.
Multiple meters	Nearly 100 meters make consumer-level clustering meaningful.
High frequency	Original measurements are at three-minute intervals.
Long period	Data spans May 2019 to October 2021, allowing repeated and seasonal behavior to be studied.
Core variable	t_kWh provides electricity consumption, the main signal for our project.
Time information	x_Timestamp lets us derive hour/day/month and weekday/weekend features.
Extra electrical variables	Voltage, current and frequency can be used as secondary features or for an extension.
Daily aggregated files	Useful for rapid prototyping before handling raw three-minute data.
Accessible	Public Kaggle copy plus Harvard Dataverse record; license is listed as CC0/Public Domain on Kaggle.

The Kaggle listing describes eight CSV files: three years each for Bareilly and Mathura, plus daily aggregated files for each city.

3. Dataset Parameters / Attributes

Attribute	Meaning	Use in our project
meter	Smart-meter identifier.	Groups all readings belonging to a meter/consumer.
x_Timestamp	Timestamp of the reading.	Creates temporal features and consumption profiles.
t_kWh	Electricity consumption in kWh.	Primary signal for profiling and clustering.
z_Avg Voltage (Volt)	Average measured voltage.	Optional secondary electrical-behavior feature.
z_Avg Current (Amp)	Average current withdrawn.	Optional secondary feature / quality analysis.
y_Freq (Hz)	Grid-frequency measurement.	Optional secondary feature / operational analysis.

4. What One Raw Record Represents

A raw row represents a measurement from one smart meter at a particular timestamp, together with electricity consumption and optional electrical measurements.

meter	x_Timestamp	t_kWh	Voltage	Current	Frequency
Bxxx	2019-05-01 10:03	0.0xx	xxx.x	x.xx	xx.xx

The numeric values above are illustrative only; the important structure is meter + time + consumption + optional electrical measurements.

5. Critical Data Transformation

The object we want to cluster is the **consumer/meter profile**, not an individual three-minute reading. Therefore, raw readings must be cleaned, aggregated and converted into meaningful per-meter features.

Stage	Action	Output
1. Clean	Handle missing values, duplicates, invalid timestamps and obvious measurement errors.	Clean time series
2. Resample	Aggregate three-minute readings to hourly or another practical interval.	Hourly/daily readings
3. Engineer features	Build consumption-level and time-of-use descriptors for each meter.	Feature matrix
4. Normalize	Scale numerical features before distance-based clustering.	Comparable feature scales
5. Mine	Run clustering and evaluate cluster quality.	Consumer profiles

6. Features We Can Derive

Feature	Derivation / meaning	Purpose
Average consumption	Mean t_{kWh} over a chosen window.	Overall usage level.
Maximum consumption	Maximum interval consumption.	Peak-demand behavior.
Minimum / lower percentile	Low-end consumption level.	Baseline usage.
Standard deviation	Variation in consumption.	Stable vs variable behavior.
Peak-hour usage	Consumption during selected high-demand hours.	Time-of-use behavior.
Off-peak usage	Consumption during off-peak hours.	Contrast with peak use.
Peak-to-average ratio	Peak / average consumption.	Sharpness of demand peaks.
Daytime share	Daytime consumption / total.	Daytime-heavy profiles.
Evening/night share	Evening/night consumption / total.	Evening-heavy profiles.
Weekday/weekend difference	Compare weekday and weekend summaries.	Schedule-driven behavior.
Monthly/seasonal features	Consumption summarized by month/season.	Seasonal behavior.

7. Algorithms We Will Use

Method	Role in project	Syllabus mapping
K-Means	Primary method for discovering consumer groups.	Experiment 9(a)
Hierarchical Agglomerative Clustering	Alternative clustering method for comparison.	Experiment 9(b)
GMM / EM	Optional probabilistic clustering extension.	Experiment 10
KNN	Classify an unseen profile using nearby profiles.	Experiment 3
Decision Tree	Classify groups and interpret important splitting features.	Experiment 4
SVM	Compare another strong supervised classifier.	Experiment 7
Logistic Regression	Optional binary/multiclass classification comparison.	Experiment 8(b)

Scope recommendation: *K-Means + HAC for clustering, followed by KNN + Decision Tree + SVM for classification. GMM and Logistic Regression can be optional extensions. We do not need every syllabus algorithm.*

8. Selecting the Number of Clusters

Measure	Interpretation
Elbow / within-cluster sum of squares	Shows where increasing k gives diminishing improvement.
Silhouette Score	Higher values generally indicate better-separated, more cohesive clusters.
Davies-Bouldin Index	Lower values generally indicate better separation/compactness.
Visual profile comparison	Checks whether clusters correspond to understandable consumption patterns.

9. From Clustering to Classification

Once K-Means produces meaningful clusters, the cluster IDs can be used as temporary class labels for a supervised-learning experiment. KNN, Decision Tree and SVM can then learn to assign a new meter profile to one of the discovered groups.

Example: Historical profiles -> K-Means -> Cluster 0/1/2 -> train KNN/DT/SVM -> classify a new profile.

These cluster labels are data-discovered consumption profiles, not predefined social categories. Their meanings should be explained only after examining the cluster characteristics.

10. Train/Test and Evaluation

Part	Plan
Training/testing	Use a time-aware or otherwise leakage-safe split when appropriate. Do not let future readings leak into training.
Classification	Accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC where applicable.
Clustering	Silhouette Score, Davies-Bouldin Index and profile visualization.
Tuning	KNN: tune K; Decision Tree: tune depth/min samples; SVM: tune kernel/C/gamma as appropriate.
Final comparison	Compare models and explain which features and methods perform well and why.

11. Expected Final Outputs

- Cleaned and transformed smart-meter dataset.
- Daily/hourly consumption profiles for each meter.
- Interpretable clusters showing distinct consumption behavior.
- Cluster-quality plots and representative profile visualizations.
- Comparison of K-Means with HAC and, if feasible, GMM.
- Classification results for KNN, Decision Tree and SVM; Logistic Regression as an optional comparison.
- Interpretation of each discovered profile and comparison of model performance.

12. Example Interpretation (Illustrative Only)

Example cluster	Possible interpretation after analysing the data
Cluster 0	Low and relatively stable consumption.
Cluster 1	Higher evening/peak-period consumption.
Cluster 2	Higher daytime usage or a different daily load shape.

These are examples only. The actual project must derive the descriptions from the measured data.

13. Direct Mapping to 23CS3551 Data Mining Lab

Syllabus item	How our project uses it
Experiment 1: Kaggle/UCI/tools	Kaggle dataset, Jupyter Notebook / Colab and Python.
Experiment 2: preprocessing	Missing values, normalization, descriptive statistics and outlier checks.
Experiment 3: KNN	Classification of unseen consumption profiles; tune K with cross-validation.
Experiment 4: Decision Tree	Classification and parameter tuning.
Experiment 7: SVM	Supervised classification and performance evaluation.
Experiment 8(b): Logistic Regression	Optional classification comparison.
Experiment 9(a): K-Means	Primary consumer-profile discovery.
Experiment 9(b): HAC	Comparison with K-Means and dendrogram visualization.
Experiment 10: GMM/EM	Optional probabilistic-clustering extension; BIC/AIC and log-likelihood.
Experiment 11: Capstone	Problem identification, dataset selection, preprocessing, algorithm selection, model building, tuning and evaluation.

14. Recommended Scope for a Beginner Team

Phase 1 - Core: Understand the raw data, clean it, create hourly/daily meter profiles, run K-Means, select k using elbow/silhouette and visualize the clusters.

Phase 2 - Strong final project: Compare K-Means with HAC, then use KNN, Decision Tree and SVM to classify unseen profiles based on the discovered groups.

Phase 3 - Optional: Add GMM/EM, extra feature engineering or a small dashboard after the core pipeline works.

The essential research question can be answered without implementing every algorithm in the syllabus.

15. Sources

Source	Link
Kaggle dataset: Electricity smart meter data from Indian two city	https://www.kaggle.com/datasets/pythonafroz/electricity-smart-meter-data-from-india
Original Harvard Dataverse record	https://doi.org/10.7910/DVN/GOCHJH
Research paper: Categorization of Indian residential consumers electrical energy consumption pattern using clustering and classification techniques (Energy, 2024)	https://doi.org/10.1016/j.energy.2023.129992
PVPSIT 23CS3551 Data Mining Lab syllabus	https://www.pvpsiddhartha.ac.in/syllabus_23/cse/31/23CS3551.pdf

Bottom line

The Mathura & Bareilly dataset is a strong choice because it combines Indian residential smart-meter data, multiple meters, high-frequency timestamps and electricity-consumption measurements. The project can be implemented entirely with concepts already present in 23CS3551, while retaining a genuine research basis.